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Product Description

GTP Proxy

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1. General Information

floLIVE's Connectivity Management Platform (CMP) is a cloud-based solution that encompasses a fully virtualized 3G/4G/LTE and 5G core network as well as a complementary suite of Business Support Systems (BSS) that enable our customers to take full ownership over their IoT business, be it device manufacturing and distribution or complete IoT connectivity management.

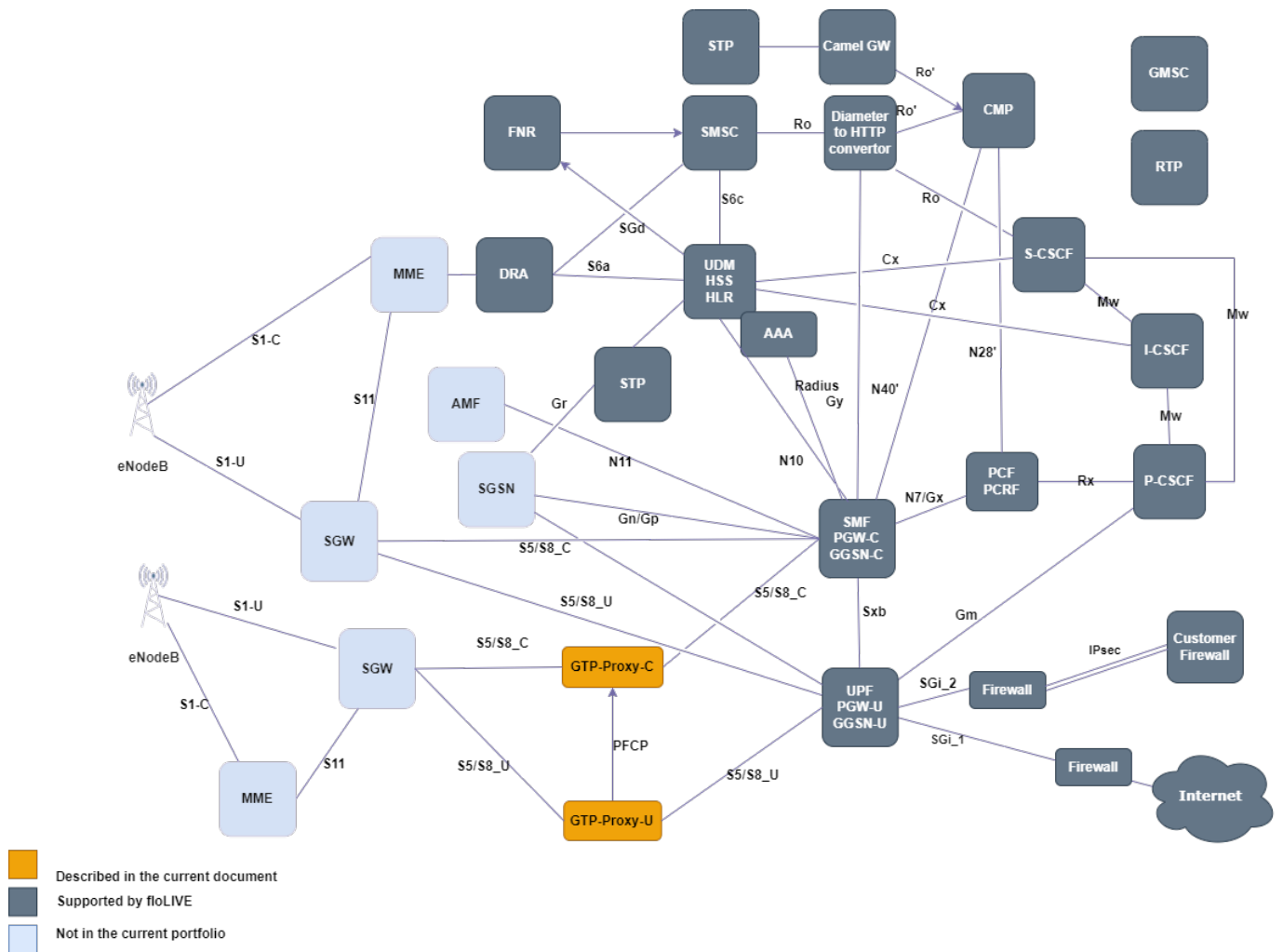
The platform provides the functionality necessary to support various workflows in managing the connectivity aspects of modern IoT and M2M device arrays.

floLIVE's CMP-as-a-Service solution comprises several subsystems and management applications.

The subsystems in use are the following:

- **floCONNECT Connectivity Management** - used for setting and managing accounts, users, SIMs, Self-care, Rules & Alerts, and more.
- **HLR/HSS** - Used for managing subscriber profiles, RAT Types, IMSI/MSISDN ranges and more.
- **GGSN/PGW (UPF & SMF)** - used for configuration and management of APNs, IP Pools, routing tables, etc.
- **SMSC** - used for configuration of SMS logic, forwarding rules and policies, external SMPP connections, and more.
- **floSIM Multi IMSI SIM Management** - used for configuration, management and control of SIM cards, business logic, PLMN definitions, OTA transactions, key management, and more.

Figure 1 - GTP Proxy



2. GTP Proxy Description

GTP proxy handles GTP V1 user plane traffic and GTP V2 control messages and allows connectivity from the mobile access network of one operator (MNO) to packet data networks of different MVNOs and CSPs.

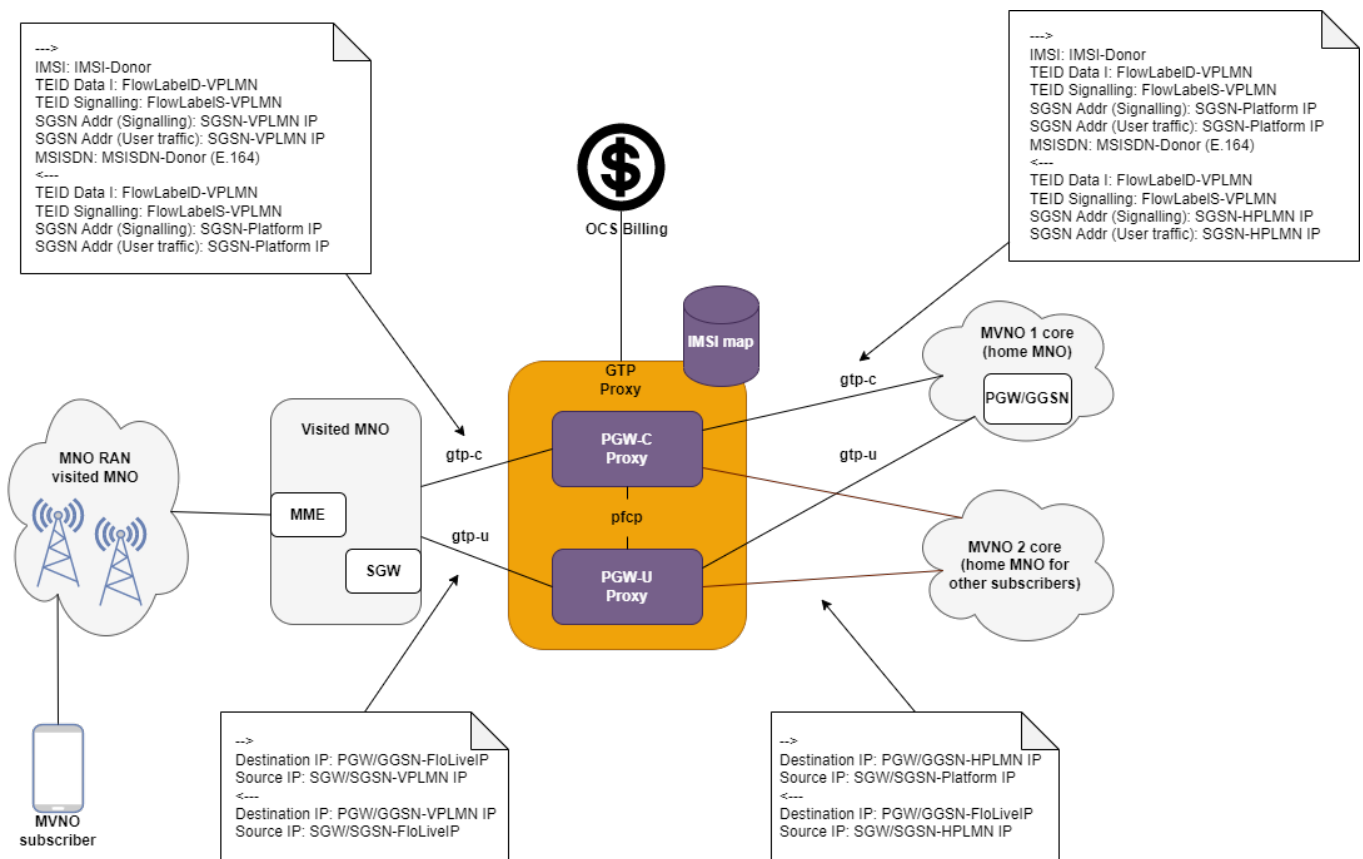
NOTE: We support both GTP V1 and GTP V2 .

MVNO selection is based on designated IMSI ranges.

The IMSI range selection is performed in the core UI.

The solution allows control messages to be modified (for example, a change of APN or IMSI).

Figure 2 - GTP Proxy Description



2.1. GTP Proxy Operation

GTP proxy terminates the GTP tunnel from RAN/MNO, examines IMSI and APN, and establishes another GTP tunnel to target/MVNO PGW. It may change APN in session establishment messages.

The user traffic is forwarded between the two tunnels.

GTP proxy measures the traffic and provides information for Billing like regular user data traffic.

3. GTP Proxy Flow

The request comes to the floLIVE GTP Proxy. The proxy must determine the type of request and which PGW it should be sent to. The IMSI can be processed in two different ways depending on the configurations.

There are two types of configurations:

- With DNS: flo LIVE Proxy sends the IMSI to DNS and the DNS determines to which PGW it should be sent.
- Without DNS: flo LIVE Proxy sends the IMSI to a PGW in which the IMSIs are hardcoded.

Figure 3 - GTP - C Proxy Flow With DNS

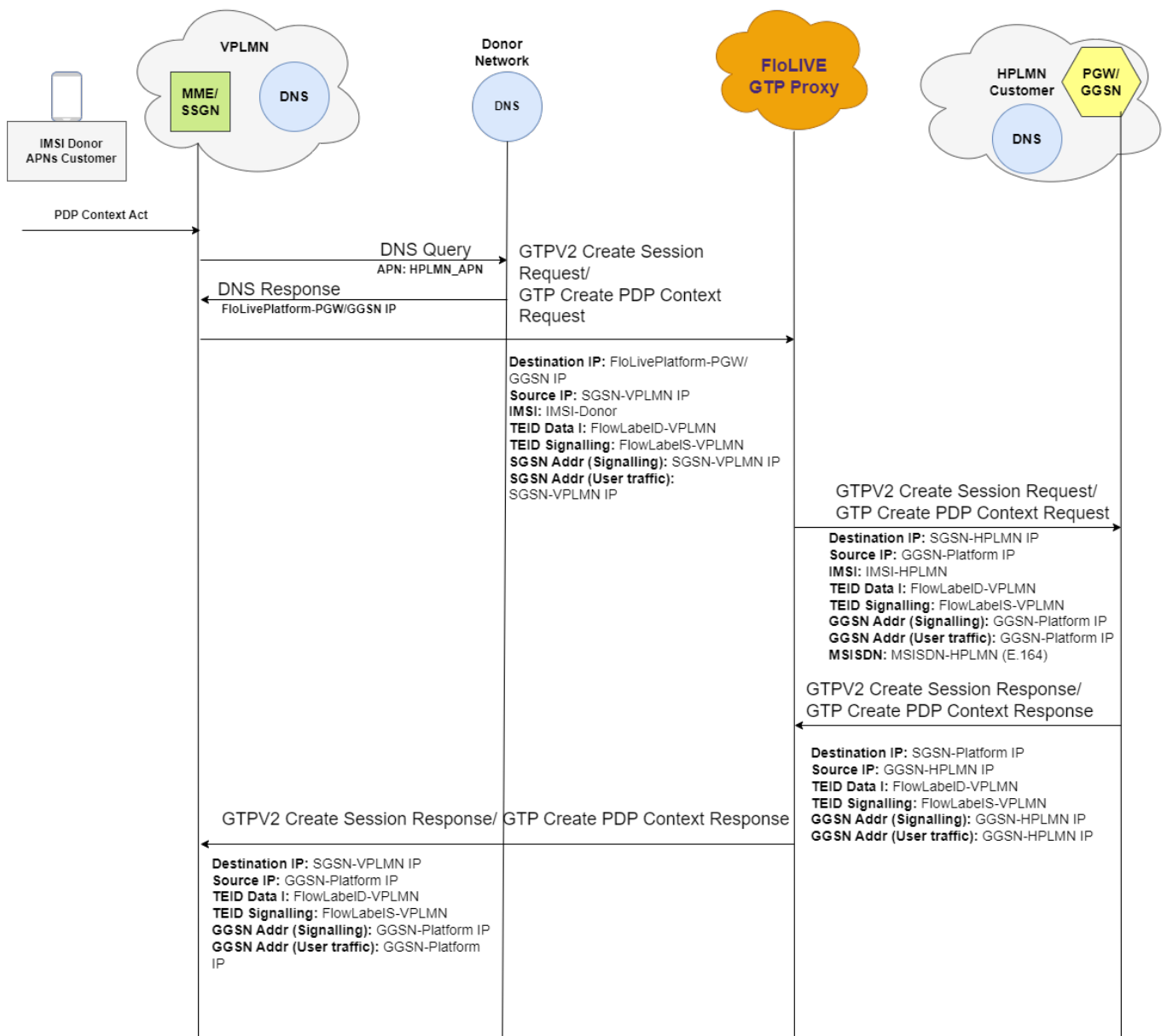
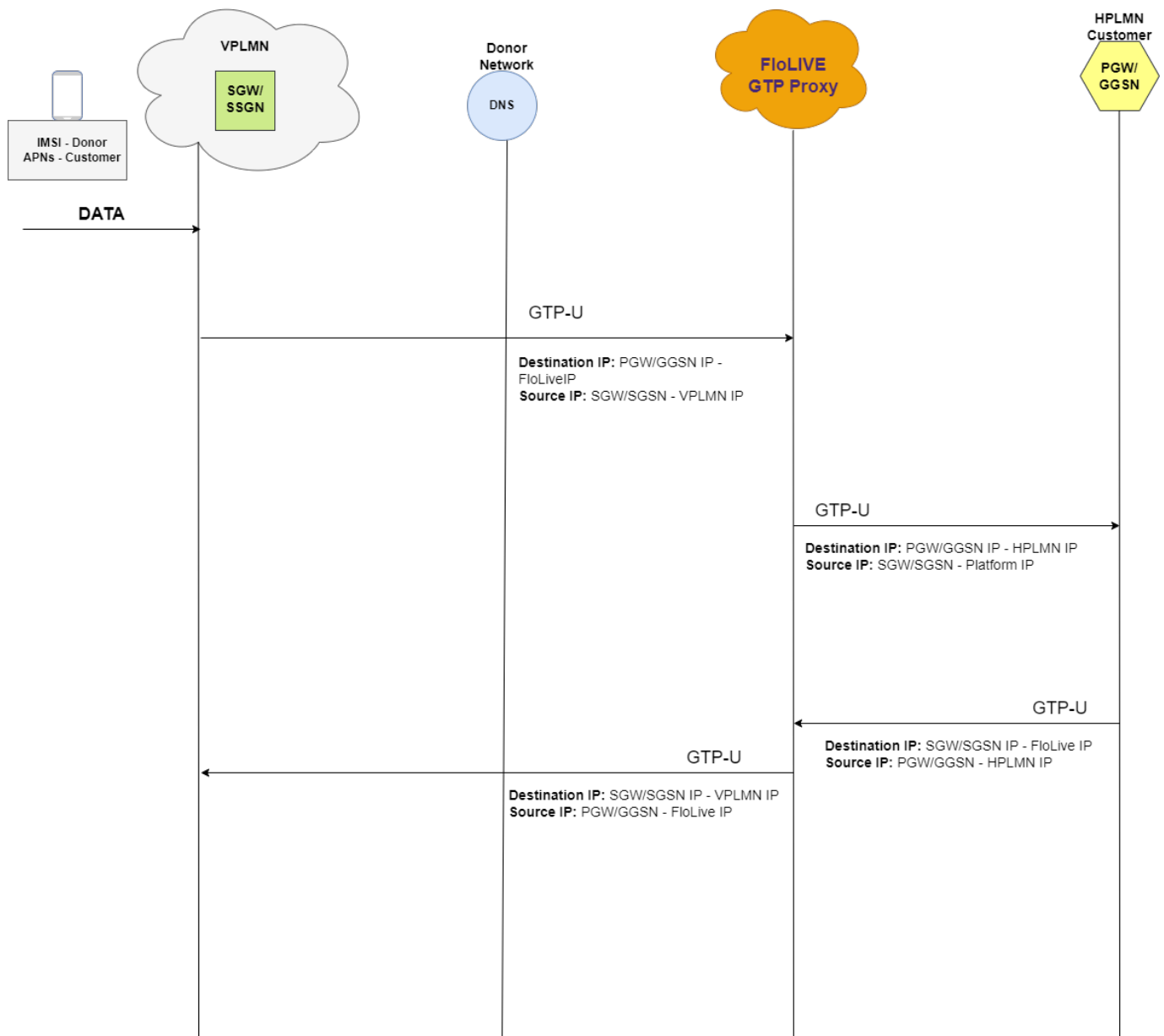


Figure 4 - GTP - U Proxy Flow With DNS



GTP Proxy – Outbound Roamer – PDP Context Activation – GRX Network (2/2)

Figure 5 - GTP - C Proxy Flow Without DNS

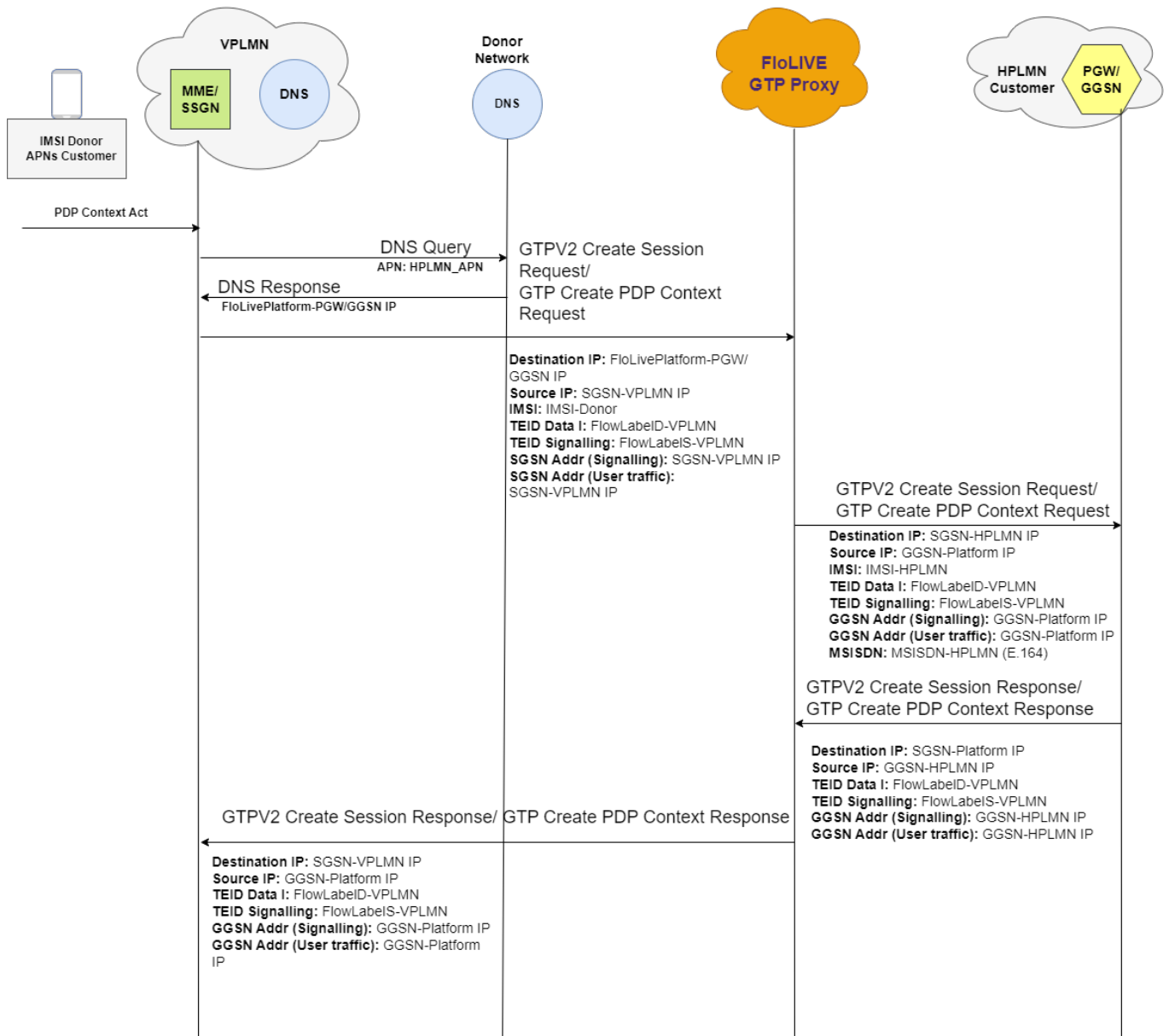
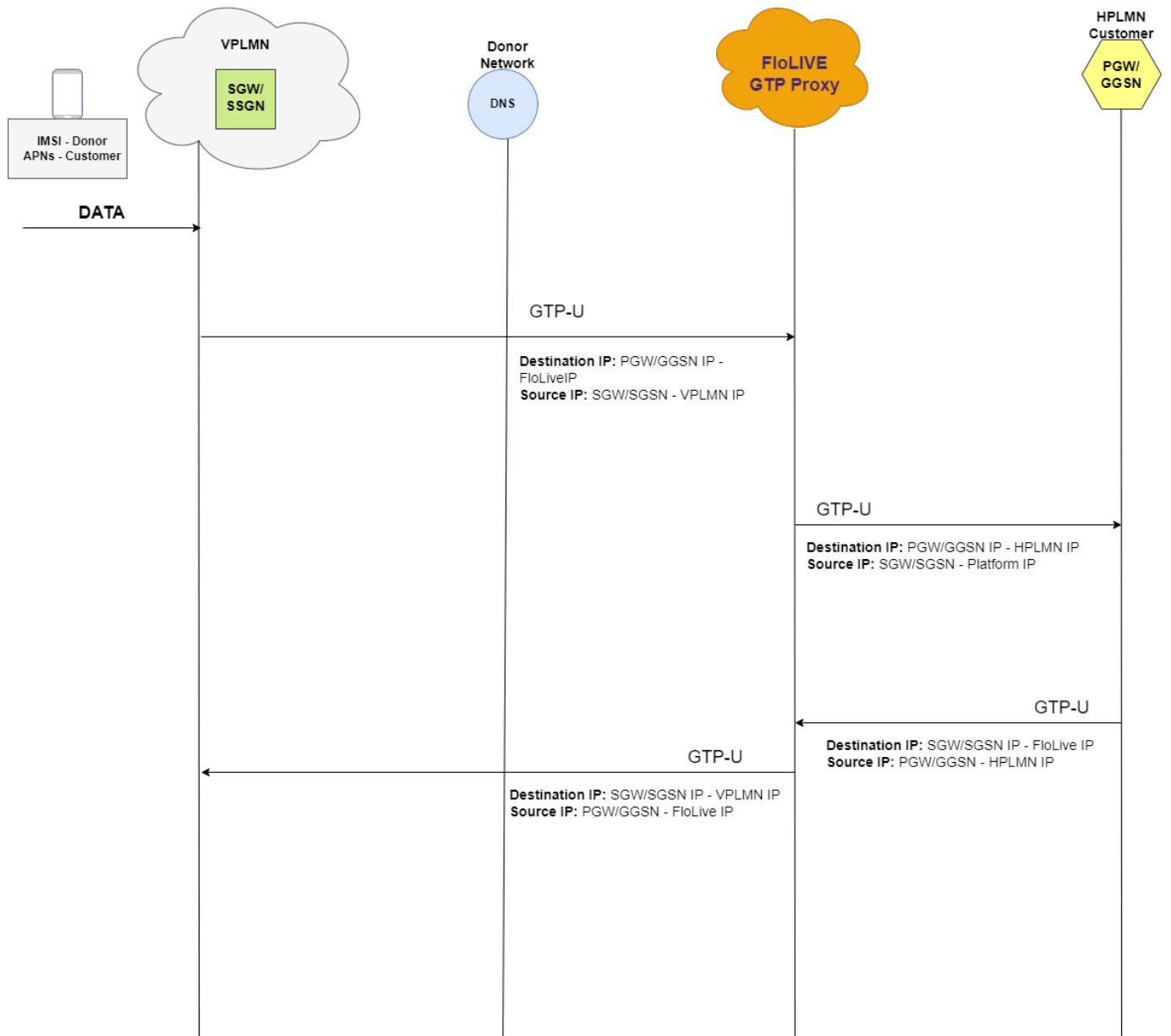


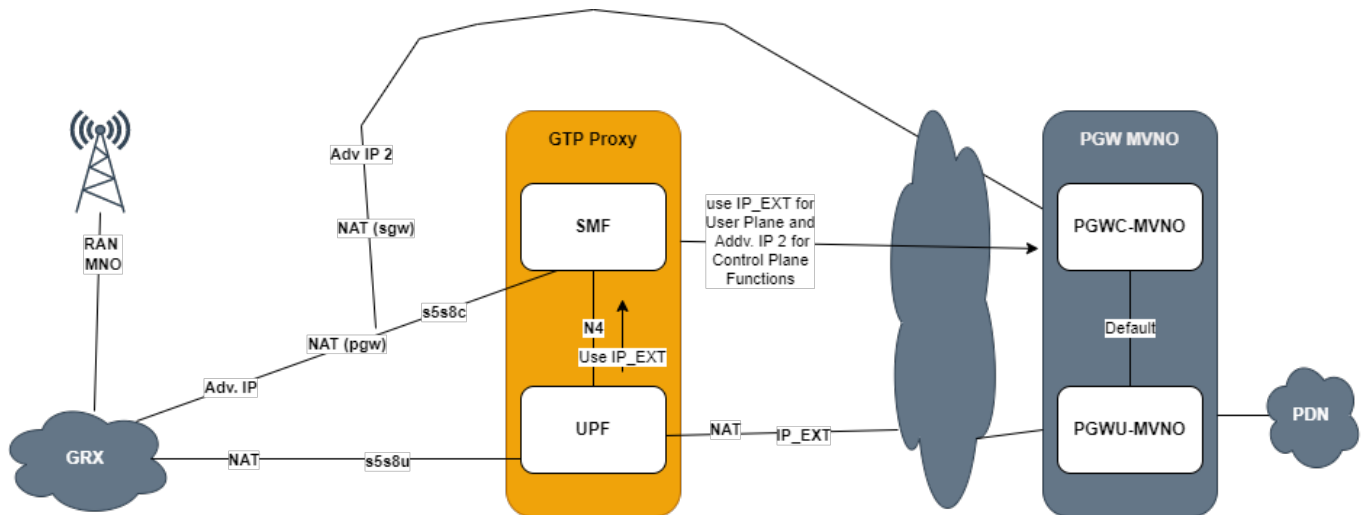
Figure 6 - GTP - U Proxy Flow Without DNS



4. GTP Proxy Integration

The diagram illustrates a GTP (GPRS Tunnelling Protocol) Proxy architecture that intermediates between an MNO (Mobile Network Operator) and an MVNO (Mobile Virtual Network Operator).

Figure 7 - GTP Proxy Integration



5. GTP Proxy Features

A GTP Proxy (GPRS Tunneling Protocol Proxy) is a control-plane and/or user-plane intermediary used in mobile core networks to manage, secure, optimize, and route GTP traffic between roaming partners, packet gateways, and mobile core elements.

5.1. GTP Signaling Relay and Routing

The proxy forwards and intelligently routes:

- GTP-C (Control Plane)
- GTP-U (User Plane)

Between:

- SGSN ↔ GGSN
- MME ↔ SGW ↔ PGW
- Visited ↔ Home networks

Typical capabilities:

- APN-based routing
- IMSI-based routing

5.2. Topology Hiding

A major roaming-security function.

The proxy masks:

- Internal IP addresses
- Core network topology
- Node identities

Benefits:

- Prevents direct exposure of EPC/GPRS elements
- Reduces attack surface
- Simplifies interconnect management

5.3. Roaming Interworking

Enables interoperability between operators and roaming hubs.

Functions:

- Protocol normalization
- Vendor interoperability
- Legacy-to-modern network adaptation

Example:

- Translating behaviors between different EPC vendor implementations.

5.4. Load Balancing

Distributes sessions across:

- PGWs
- GGSNs
- UPFs (in some 5G-interworking scenarios)

Methods:

- Session-aware balancing
- Subscriber-aware balancing
- Geographic balancing

Benefits:

- Scalability
- High availability
- Improved resource utilization

5.5. Charging and Policy Integration

Can interact with:

- PCRF/PCF
- OCS/OFCS
- Charging gateways

Functions:

- Policy enforcement
- APN restrictions
- Usage tracking

5.6. Traffic Steering

Routes traffic based on:

- APN
- Subscriber profile
- Geography
- Roaming partner
- Service type

Examples:

- IoT breakout
- Enterprise APN steering
- Local breakout (LBO)
- Home-routed roaming

5.7. Lawful Intercept Support

Some implementations support:

- LI mediation
- Regulatory interception workflows
- Subscriber trace functions